

Adaptive Self-Consistency for Long-Form Factuality on Long-Tail Facts

Aggregation, saturation, and budget-aware inference

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Fluent long answers can hide many factual failures

The setting

- Entity-centric questions require **long, coherent answers**.
- Each answer contains many independently checkable claims.
- Rare entities make parametric recall less reliable.

A convincing answer
many atomic claims



Unsupported claims
remain persuasive

long tail

Fluency $\neq$ factual precision

Core tension

More samples can reveal correct facts. A fixed large budget still wastes compute after gains level off.

Three questions organize the study

1

Does sampling help?

Across head, torso, and tail entities.

2

How should candidates combine?

Synthesis versus voting and selection.

3

Can compute be routed?

Preserve factuality while reducing samples.

Empirical thesis

LLMAgg provides the main factuality gain; adaptive routing compresses a conservative budget, but does not replace calibration.

Evaluation: long-form facts across popularity tiers

RiDiC benchmark

- 3 domains: rivers, cars, disasters.
- 2 languages: English and Chinese.
- Wikipedia pageview tiers: **head**, torso, **tail**.
- Public pool: up to 99 completions per input.

Fact-level metric

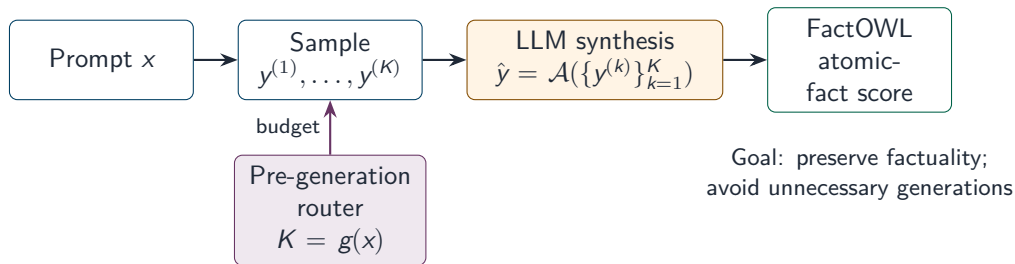
$$F(y) = \{f_1, \dots, f_n\}$$
$$\text{Fact}(y) = \frac{1}{|F(y)|} \sum_{i=1}^n \mathbf{1}[f_i \text{ supported}]$$

FactOWL extracts atomic facts and verifies them against the supplied Wikipedia context.

Same generator: Llama 3.1 8B Instruct

Lang.	Domain	Tier	RiDiC entity example
EN	cars	tail	Jiotto Caspita
EN	rivers	head	Blue Nile
EN	disasters	torso	Super Outbreak
ZH	cars	tail	劳斯莱斯 Silver Wraith
ZH	disasters	torso	2024 年花莲地震
ZH	rivers	head	伏尔塔瓦河

Method: sample, synthesize, score, then route the budget



- **Single:** one completion, no aggregation.
- **LLMAgg:** merge consistent claims and remove contradictions.
- **Adaptive:** choose K before any candidate is generated.

Why synthesis can beat selecting one answer

	Candidate A	Candidate B	Candidate C
Claim 1	SUV / MG ZS platform	electric SUV	all-electric sedan
Claim 2	1.5L engine	range up to 600 km	unveiled in 2023
Claim 3	manual transmission	Tesla Model Y	600+ horsepower variant

Example: Xiaomi SU7; Candidate B also mentions a 2022 launch.

Aggregation objective

Preserve supported, complementary facts; omit unsupported claims and contradictions.

- Selection can only inherit the errors of one existing response.
- LLMAgg writes a new response from the pooled evidence.

Baselines separate agreement, selection, and synthesis

Condition	What it does	Deployable?
Single	One independently sampled completion	yes
MC	Majority co-occurrence over responses	yes
MF	Majority voting over normalized atomic facts	yes
SP (SeqProb)	Select highest mean token log-probability	yes
Best-of- K logprob	Same selector on the matched $K = 20$ pool	yes
Best sampled	Oracle picks factuality-best candidate after scoring	no
LLMAgg	LLM synthesizes a new response from the pool	yes

Fair comparison

The compute-matched best-of- K selector uses the same 20-sample pool as LLMAgg@20; full-budget results use 99 candidates.

LLM synthesis is the strongest deployable condition

	Single	Best samp.*	MC	MF	SP	LLMAgg@99
English macro	0.279	0.549	0.128	0.115	0.332	0.578
Chinese macro	0.322	0.634	0.201	0.028	0.394	0.571

* Oracle diagnostic: selected after scoring; not deployable.

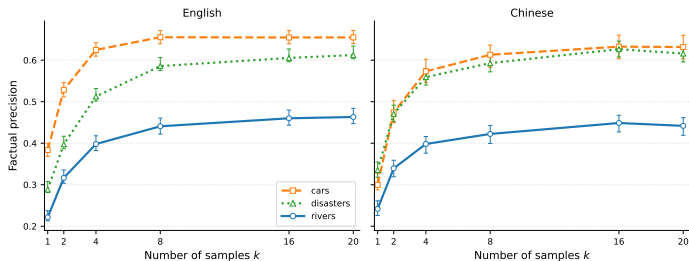
Across six language-domain cells

- LLMAgg is strongest among deployable methods.
- Gains are broad, not driven by one domain.

Interpretation

Synthesis consolidates partially correct material;
next: how quickly does that gain saturate?

Budget gains level off around $K = 8$ to $K = 16$, with domain exceptions



What the curves show

- Sharp gains at small K .
- Saturates by roughly $K = 8$ to $K = 16$ in many cells.
- EN disasters improves through $K = 20$.
- ZH rivers is non-monotonic.

Nested prefixes from one fixed 99-candidate pool; no resampling at each point.

Aggregation improves tail entities, but the head-to-tail gap remains

Lang.	Tier	Single	LLMAgg@20
		macro Fact	macro Fact
EN	head	0.445	0.828
EN	torso	0.384	0.762
EN	tail	0.243	0.513
ZH	head	0.480	0.757
ZH	torso	0.396	0.675
ZH	tail	0.275	0.501

Stable pattern

Head > torso > tail remains visible after improved inference.

Implication

Aggregation raises precision broadly; long-tail difficulty remains.

Values are macro averages over the three domains within each language and tier.

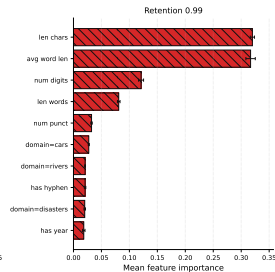
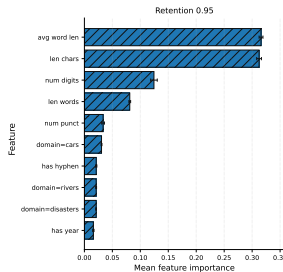
Adaptive routing uses pre-generation complexity signals

Controller

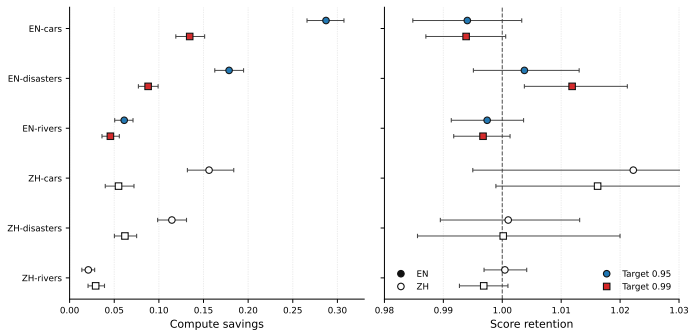
Random forest predicts a budget $K = g(x)$ before generation.

- 300 trees; minimum leaf size 3.
- Prompt length, lexical complexity, punctuation, years.
- Language/domain indicators and a domain budget prior.
- No partial output or disagreement features.

Length + average word length account for over 60% of feature importance.



Against $K_{\text{full}} = 99$: observed savings, retention stays near 1



Main operating result

2–29%

sampled-generation savings

Retention targets

- Target 0.95: retention 0.994–1.022.
- Target 0.99: retention 0.994–1.016.
- Coverage varies by domain and target.

Smaller routed budgets can occasionally score above the full-budget reference because aggregation is not strictly monotonic.

Adaptive routing saves against $K = 99$, not $K = 20$

$K = 20$ is a different reference

- Budget curves identify $K = 20$ as near-saturated.
- Against $K = 20$, the controller produces no net savings.
- Predicted budgets average 70.6–96.9 samples.

Savings depend on the reference budget.

Generalization and safety

- Held-out-domain coverage drops to roughly 0.24–0.65.
- Routing does not detect inputs where LLMagg degrades.
- One generator family; evaluator noise remains possible.

Same-distribution routing, not a domain-agnostic recipe.

Three takeaways

① **Synthesis beats selection for long-form factual precision.**

The main gain comes from consolidating complementary facts, not choosing a single “best” completion.

② **Sampling gains saturate early, but saturation is heterogeneous.**

Domain and language determine how much additional computation remains useful.

③ **Adaptive routing reduces the $K = 99$ budget; it does not beat $K = 20$.**

It saves 2–29% against 99 samples, while $K = 20$ sensitivity and failure detection remain open.

Answer to the opening tension

Better consolidation matters more than simply generating more answers.

References

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Self-Consistency Improves Chain of Thought Reasoning in Language Models. ICLR, 2023.

 [J. Kim et al.](#)
Reliability-Aware Adaptive Self-Consistency for Efficient Sampling in LLM Reasoning. ACL Findings, 2026.

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FactOWL: A Cost-Efficient Tool for Long-Form Factuality Evaluation. SIGIR, 2026.

Thank you

Questions?

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Backup slides

Additional tables, definitions, and failure cases

Main comparison by language and domain

Lang.	Domain	Single	LLMAgg@99	Gain
EN	rivers	0.211	0.462	+0.251
EN	cars	0.344	0.653	+0.309
EN	disasters	0.283	0.620	+0.337
ZH	rivers	0.266	0.451	+0.185
ZH	cars	0.344	0.632	+0.288
ZH	disasters	0.355	0.631	+0.276

Reading the table

LLMAgg is the strongest deployable method in every language–domain cell; the oracle best-sampled diagnostic is intentionally excluded from this table.

Supplementary baselines at matched $K = 20$

Lang.	Domain	Single	Best-of- K logprob	ReASC	LLMAgg@20
EN	cars	0.344	0.366	0.354	0.655
EN	disasters	0.287	0.324	0.305	0.612
EN	rivers	0.211	0.221	0.210	0.463
ZH	cars	0.349	0.407	0.349	0.633
ZH	disasters	0.354	0.417	0.368	0.616
ZH	rivers	0.269	0.302	0.277	0.442

Audit note

ReASC is the corrected post-hoc baseline: completion-level FactOWL scores replace the earlier topic-level join.

Best-of- K logprob and ReASC are compared with LLMAgg at the same $K = 20$ budget.

Adaptive policy: definitions and training protocol

Reported quantities

$$\text{compute savings} = 1 - \frac{\sum_i K_i^{\text{adaptive}}}{N K_{\text{full}}}$$

$$\text{score retention} = \frac{\text{Fact}(\hat{y}_{\text{adaptive}})}{\text{Fact}(\hat{y}_{\text{full}})}$$

$$\text{coverage} = \Pr[\text{retention meets target}]$$

Protocol

- Five stratified shuffle splits.
- Test fraction: 30%.
- Split seed: 13.
- Model seed: 13 + fold index.
- Targets: 0.95 and 0.99.

The reported savings count sampled generations; aggregation prompt growth is not included in that relative measure.

Dataset filtering and FactOWL details

Usable-context counts

	rivers	cars	disasters
English	1000	997	1000
Chinese	723	400	723

Filtering requires usable entity titles and supplied Wikipedia contexts.

FactOWL score

$$F(y) = \{f_1, \dots, f_n\}$$
$$\text{Fact}(y) = \frac{1}{n} \sum_i 1[f_i \text{ supported}]$$

Verification uses the entity's supplied Wikipedia page; no alternative evidence corpus is queried.

Qualitative behavior: complementary evidence versus correlated errors

Representative wins

- Jiotto Caspita: +0.415.
- Blue Nile: +0.424.
- Ma Yimoka River: +0.522.

Correct facts are fragmented across candidates; synthesis recovers the shared core.

Representative failures

- Hurricane Debby: −0.548.
- Hualien earthquake 2024: −0.740.
- Vltava River: −0.556.

Repeated hallucinations or entity ambiguity can create a confident wrong consensus.

More samples help only when they add recoverable factual signal.